Ensure to carefully follow the instructions on *code_solution.pdf* to reproduce the entire procedure.

1. Combining the feedback spreadsheets into a single clean dataset

1.1 Understanding the general context and information of the feedback

Focus: customer experience i.e. feedback, reviews and comments to reflect dealership performance

Observations & Assumptions after brief analysis of the spreadsheets:

- Ratings seem to be formatted very differently across many different spreadsheets
- A *Comment* complements its respective *Rating/Score* from a customer and vice-versa
- There are missing values which make the datasets unclean
- The provided feedback is most likely not be the entire Population of the Kestrel's customer base and subsequently won't reflect the overall sentiment since only those who chose to provide feedback are recorded as data entries in the spreadsheets.

Spreadsheet characteristics:

- Each row is a feedback from a customer
- The formatting of identical columns differ between different spreadsheets
- With 52 spreadsheets on feedback, an appropriate AI LLM, Claude is essential to partially automate the process of merging the spreadsheets together within the given time period.
    - Claude Sonnet (Medium - free tier) can only accept 20 files max. at a time. So, in order to merge all the feedback spreadsheets together, they need to be separated into 3 separate folders with the following: 20, 20, 12 spreadsheets within three folders named: feedback_{1,2,3}

      Note: Any AI LLM tool used for automation may perform some unexpected operations on the data when it comes to data cleaning which may not be explicitly stated by the user. However, some prompts below have made clear to Claude not to make any unexpected or highly transformative changes.

Data Quality Rules for the Single, Cleaned Dataset:

- **[SUCCESS]** *Date* was formatted as 'YYYY-MM-DD' consistently across all rows using Python (with the help of Claude)
- **[SUCCESS]** *Rating/Score* was formatted on a consistent (1 – 5) scale using Python
- **[SUCCESS]** Duplicate rows were deleted using Python
- **[SUCCESS]** Rows with empty entries of interest (*Rating/Score* and *Comment*) for insights should be deleted using Excel

1.2 Data Merging

Task 1: Get Claude to check what common columns/attributes appear in all 52 spreadsheets

Prompt 1: *Go through all these spreadsheets and list for me the columns (based on identical meaning) that appear in all spreadsheets*

Note: Uploaded all spreadsheets from feedback's 1 to 3 folders separately per prompt

Results: Four common columns across all 52 spreadsheets and 3 feedback folders (feedback_{1, 2 & 3})

→ Date, Department, Rating/Score, Comment

Task 2: Get Claude to merge all 52 datasets by row based on the four common columns above as well as noting the *Suburb* name from the file name and adding that to a new column.

Prompt 2: *Merge all these spreadsheets together into a single file called merged_feedback_{1,2,3}.csv by stacking on all rows based on the 4 columns: Date, Department, Rating/Score and Comment (take into consideration different formatting of column names). Also, take the respective suburb name from the name of each separate spreadsheet and add them to a newly created column called Suburb.*
Finally, provide me the Python code used to perform the operations.

Note: Upload the spreadsheets from feedback's 1 to 3 folders separately per prompt

Output: Three separate *merged_feedback_x.csv* files in the *combined_datasets* folder

Task 3: Get Claude to merge the three *merged_feedback_x.csv* files together by stacking on the rows

Prompt 3: *Merge these three .csv files into a single file called merged_all_feedback.csv by stacking on all rows.*
Then provide me the Python code used to perform the operations.

Output: A single merged unclean dataset: *merged_all_feedback.csv* in the combined_datasets folder

1.3 Data Cleaning

Task 4: Get Claude to analyse the rows of *merged_all_feedback.csv* for any inconsistent formatting and report on them. With Rating/Score being a Categorical Ordinal variable seems to be significant for this task which requires consistent formatting. Whereas, Comment is human textual input which cannot be formatted consistently and would require some sentiment analysis.
Then according to the output, get Claude to fix the formatting for consistency then find any duplicate rows and remove them.

Prompt 4: *Take the following spreadsheet and look for and report on any inconsistent formatting across the rows. Pay close attention to the Rating/Score column. Do not make any changes to the data!*

Summary of Response:

- *Inconsistent formatting in the 4 common columns with the Rating/Score column being the worst*
- *There are case, formatting and spelling inconsistencies in Department names*
- *24% of rows have a blank Comment*
- *There are 198 duplicate rows*
- *Suburb is completely clean*

Prompt: *Ok, take all the formatting issues you've found and fix them. Also, delete any duplicate rows. Then output a file called merged_all_feedback_cleaned.csv*
Finally, provide me with the Python code used to perform the operation.

Output: A single merged cleaner dataset: *merged_all_feedback_cleaned.csv*in the *combined_datasets* folder

Extension and Discovered Insight: There can be no customer experience insight derived from a customer (as a data row) who provides no Rating or Comment. So get Claude to find any rows that don't have a Rating or Comment.

Task 5: Get Claude to look through *merged_all_feedback_cleaned.csv* and find the rows that don't have a Rating/Score AND no Comment.

Prompt 5: *Did any rows have an empty Rating AND empty comment?*

Response: Yes, two rows from the Tamworth region

Manual Intervention: Opened *merged_all_feedback_cleaned.csv* in Excel to found them by filtering on the Suburb column for Tamworth and deleted them. This should not have

any impact on the results for the following section given the Sample Size of the single combined dataset is around 7000 data entries.

Result: A single merged cleaner dataset: *merged_all_feedback_cleaned.csv* in the *combined_datasets* folder

- **Incorrect Format Discovered:** The *Suburb*, 'North' is comprised of 2 *Suburbs*: North QLD Rocky and North QLD TSV based on the file name in the *feedback_2* folder.
  As a result 23 Dealership Suburbs are part of below analysis instead of 24.

Final Combined Dataset:
combined_datasets/merged_all_feedback_cleaned.csv
(with 7041 rows by 5 columns)

Data Dictionary

Column	**Definition**	**Data Type**	**Range/Categories**
Date	Date the feedback was made by a customer	Date (YYYY-MM-DD)	2025-10-1 to 2026-09-30
Suburb	Suburb/Region of a Dealership	Categorical Nominal	23 Dealership Suburbs (*see Pivot Table below*)
Department	The different business department that deal with different customer products/services	Categorical Nominal	Parts, Ratings, Services
Rating/Score	Feedback Rating/ Score	Categorical Ordinal	1 to 5
Comment	The customer's written feedback	Free form Text	-
Sentiment for Comment *(only in SentAnaly.csv file)*	Classification on whether a *Comment* evokes a Negative, Okay or Positive sentiment overall	Categorical Ordinal	Positive, Okay, Negative

2. Data & Statistical Analysis of Ratings

To answer the following business questions:

- What are the main drivers of negative feedback, and how do they vary across dealerships and departments?
  - Pie Charts with Pivot Tables were created
- Has anything changed over the year?
  - Time Series of Ratings were created from Pivot Tables

- Where should Kestrel focus first, and roughly what is it worth?
    - Identify areas (Dealership Suburbs and Departments) to target improvement where 1 and 2 ratings are made

Note: Negative Feedback (main focus) has been classified as a 1 or 2 out of 5 Rating and Positive Feedback has been classified as 4 or 5 out of 5 Rating

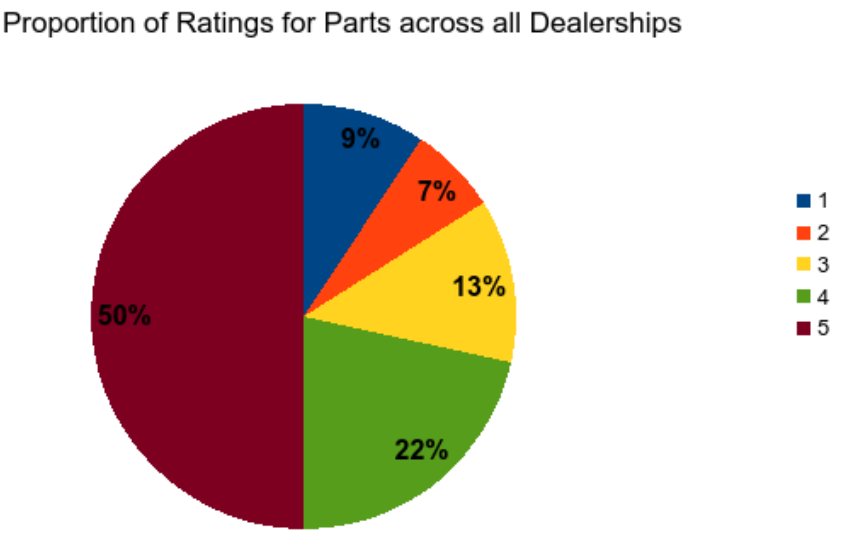
***merged_all_feedback_cleaned.csv* was opened in Excel and a copy was saved as a .xlsx file within the root directory**

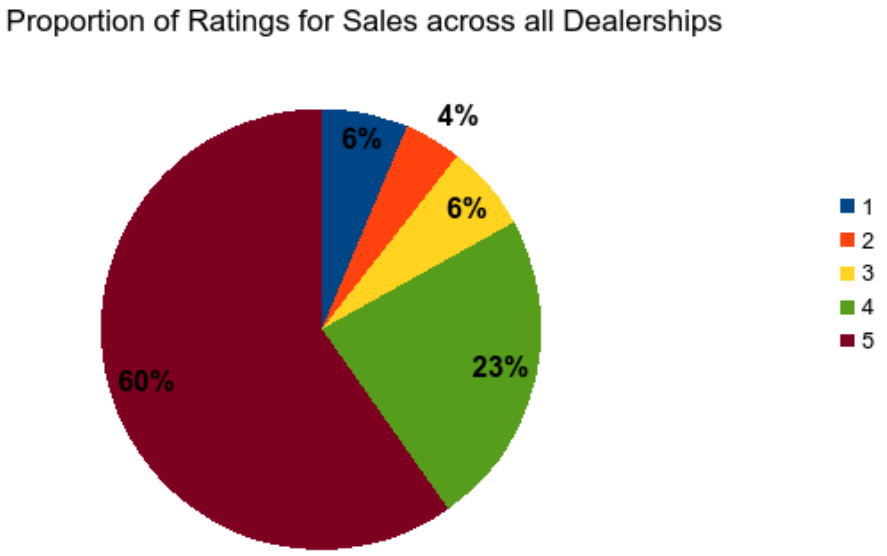
2.1 Pivot Table in Excel to summarise the Ratings/Scores across Dealership Suburbs and Departments

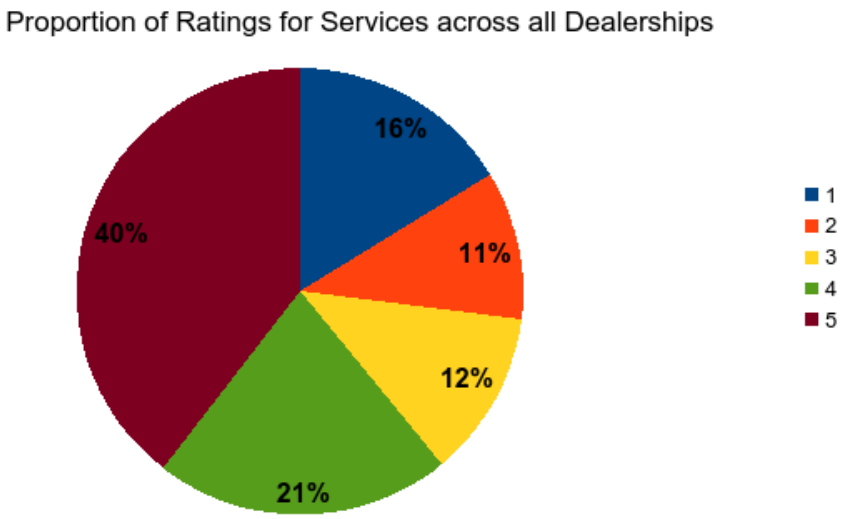
Count - Ratin	Departme	Rating/Scor															
	Parts						Sales					Service					Total Result
Suburb	1	2	3	4	5		1	2	3	4	5	1	2	3	4	5	(empty)	
Ballarat	4	4	4	7	18		3	2	3	20	31	28	24	27	36	79		**290**
Bendigo	1	2	1	4	7			1	3	11	17	19	6	10	26	36		**144**
Box Hill	4		3	9	13		10	6	7	18	36	30	24	27	54	66		**307**
Chatswood	1	2	3	12	14		7	3	4	13	39	31	25	17	39	68		**278**
Chermside	3	5	5	5	19		8	6	10	27	58	48	39	48	64	113		**458**
Dandenong	4	3	6	10	26		4	2	7	27	74	68	42	49	51	103		**476**
Essendon	2	4	3	7	14		7	4	7	20	52	46	22	32	47	76		**343**
Frankston	2	5	6	2	15		5	3	4	14	40	32	25	22	49	82		**306**
Geelong	1	4	7	6	13		3	2	3	18	43	34	25	23	39	83		**304**
Hunter	6	1	4	4	15		3	4	5	22	34	35	18	20	38	91		**300**
Ipswich	2		3	5	16		3	3	5	13	35	28	25	12	47	80		**277**
Liverpool	3			8	14		8	6	11	17	37	28	22	16	43	96		**309**
Noosa				2	4					2	14	1			10	16		**49**
North	6	8	8	21	34			1	3	6	5	70	47	57	105	183		**554**
Parramatta	6		7	11	22		8	6	5	16	86	70	43	49	41	102		**472**
Penrith	7	4	3	13	27		11	5	3	22	73	45	26	33	77	101		**450**
Ringwood	3	1	6	8	17			2	4	26	78	34	27	36	82	129		**453**
Shepparton			2		4				1	1	9	7	1	3	9	15		**52**
Southport	1	2	3	4	8		10	5	5	16	32	27	18	21	35	61		**248**
Springwood	2		3	7	17		2	1	2	20	49	29	17	14	47	88		**298**
Tamworth	1				4		4		2		26	16		17		63		**133**
Toowoomba	2		4	2	18					3	6	29	27	23	50	84		**248**
Wollongong	4	1	6	3	8		1	2	4	23	41	26	13	25	42	85		**284**
Total Result	**65**	**46**	**87**	**150**	**347**		**97**	**64**	**98**	**355**	**915**	**781**	**516**	**581**	**1031**	**1900**		**7033**

from merged_all_feedback_cleaned.xlsx → Pivot Table – Focus on Departments (Sheet)

2.2 Pie Charts to convey the proportion of the 1 to 5 Ratings for different Departments







from merged_all_feedback_cleaned.xlsx → Proportion of Negative Ratings

Approximate (%)	**Negative (1 or 2) Ratings**	**Positive (4 or 5) Ratings**
Parts	16	72
Sales	10	83
Services	27	61

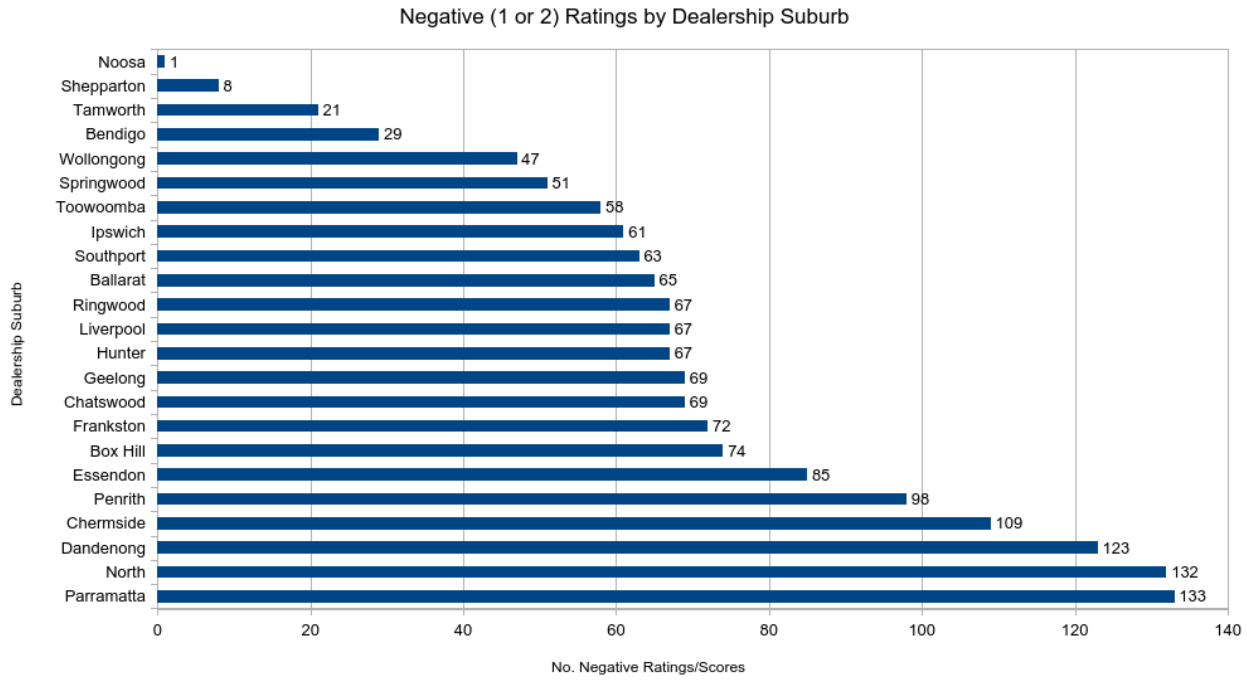
Analysis
With Services having the highest proportion of Negative ratings and lowest proportion of Positive ratings, this Department has the poorest performance compared to Parts and Services.

2.3 Pivot Table in Excel to summarise the Ratings/Scores across the Dealership Suburbs

Count - Rating/Score	Suburb																							
Rating/Score	Ballarat	Bendigo	Box Hill	Chatswood	Chermside	Dandenong	Essendon	Frankston	Geelong	Hunter	Ipswich	Liverpool	Noosa	North	Parramatta	Penrith	Ringwood	Shepparton	Southport	Springwood	Tamworth	Toowoomba	Wollongong	Total Result
1	35	20	44	39	59	76	55	39	38	44	33	39	1	76	84	63	37	7	38	33	21	31	31	943
2	30	9	30	30	50	47	30	33	31	23	28	28		56	49	35	30	1	25	18		27	16	626
3	34	14	37	24	63	62	42	32	33	29	20	27		68	61	39	46	6	29	19	19	27	35	766
4	63	41	81	64	96	88	74	65	63	64	65	68	14	132	68	112	116	10	55	74		55	68	1536
5	128	60	115	121	190	203	142	137	139	140	131	147	34	222	210	201	224	28	101	154	93	108	134	3162
(empty)																								
Total Result	290	144	307	278	458	476	343	306	304	300	277	309	49	554	472	450	453	52	248	298	133	248	284	7033

from merged_all_feedback_cleaned.xlsx → Pivot Table – Focus on Dealership Suburbs (Sheet)

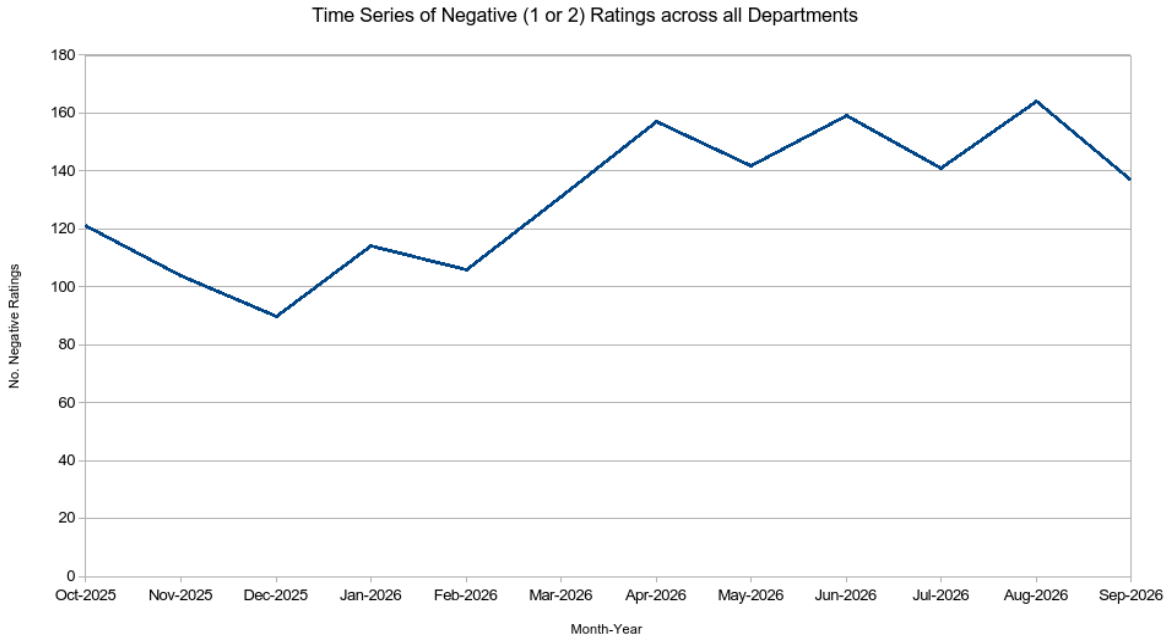
2.4 Bar Chart to show the number of Negative Ratings across Dealership Suburbs



from merged_all_feedback_cleaned.xlsx → Pivot Table - Focus on Dealership Suburbs

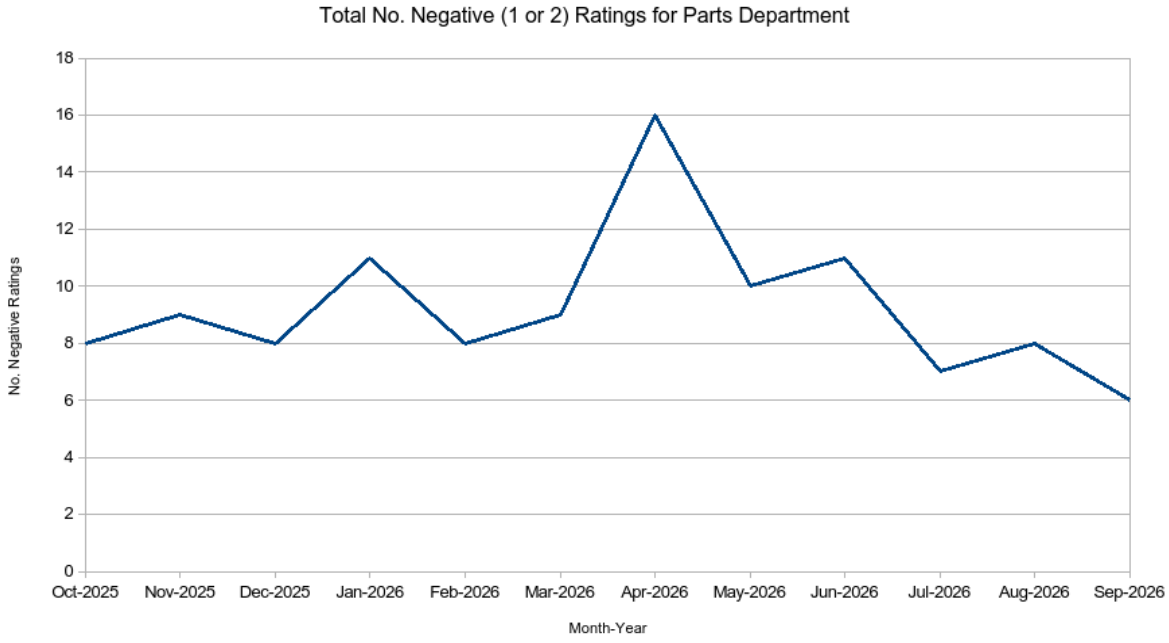
- Given that ‘North’ is already comprised of two Dealership Suburbs, ‘North QLD Rocky’ and ‘North QLD TSV’, it is expected that it would contain the second most amount of negative Ratings making it an anomaly. Conversely, Noosa contained a lot of missing Rating data which also makes its number of negative Ratings extremely low.
    - Single *Dealership Suburbs* with the most negative ratings overall: Parramatta, Dandenong, Chermside and Penrith which should have improvement processes directed to them.
    - Customers at the Noosa Dealership should have more feedback surveys sent to them or there may be a technical issue regarding feedback data collection with that specific Dealership.
- Median Negative Ratings between the 23 Dealership Suburbs: 67

2.5 Time Series of Negative Ratings over time

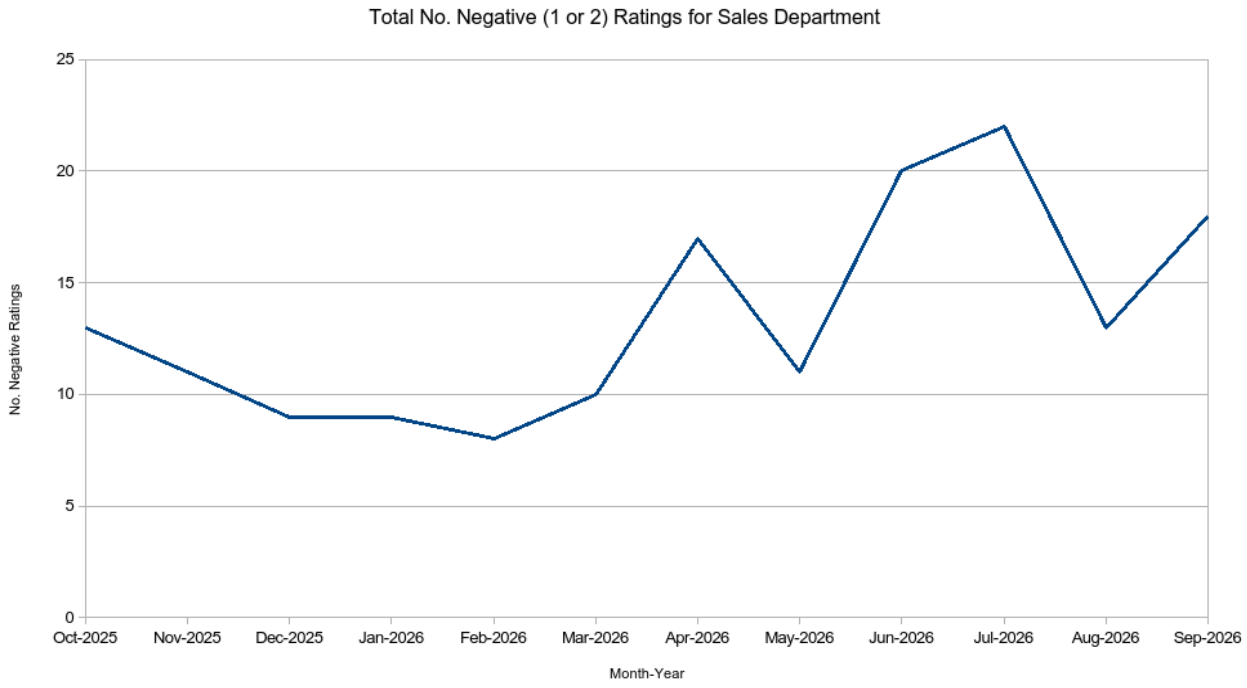


from merged_all_feedback_cleaned.xlsx → Time Series - Negative Ratings for all Departments (Sheet)

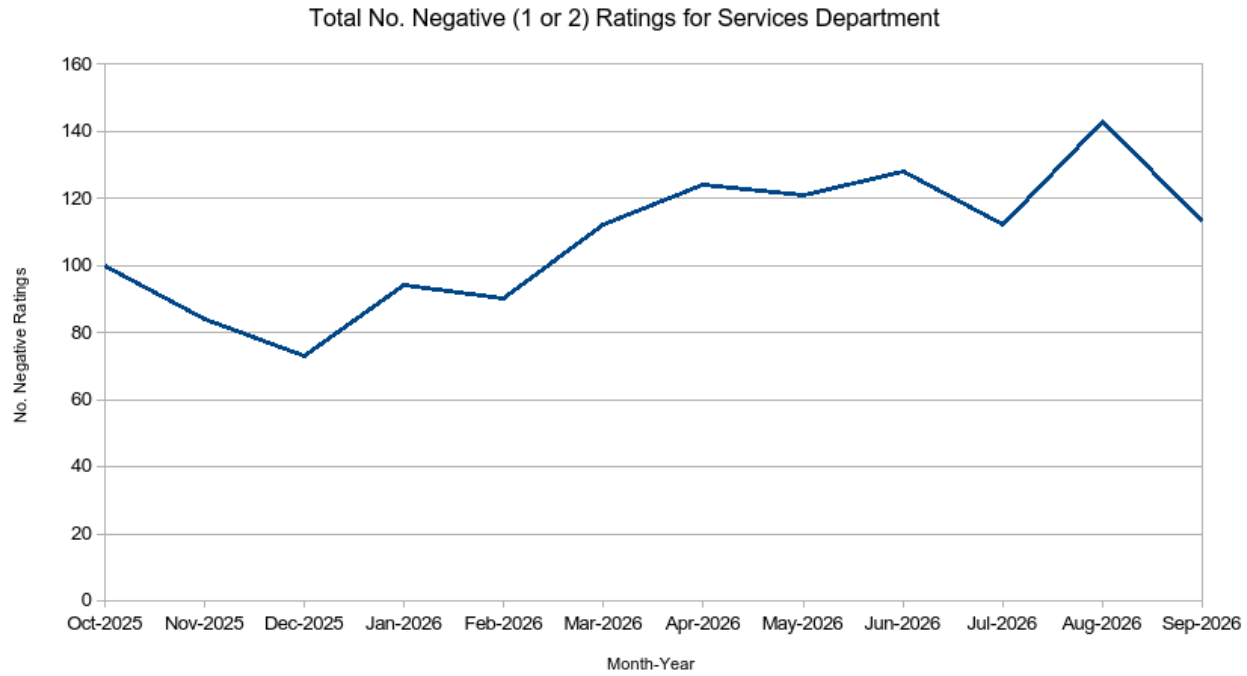
- Across all three *Departments*, the number of negative ratings fell to its lowest point of 90 in Dec 2025 and peaked at 164 during Aug. 2026
- The biggest rise was from Feb. 2026 to Apr. 2026 where the number of negative *Ratings* was above 200
- Overall, negative *Ratings* are on the rise throughout most of 2026



- Total negative *Ratings* for the Parts *Department* remained fairly consistent from Oct. 2025 to Mar. 2025 until it peaked at 16 during Apr. 2026 then a downward trend to its lowest point during Sep. 2026 of just 6.



- Total negative *Ratings* for the Sales *Department* underwent a slow decline until Feb. 2026 but was on the rise for most of 2026 with some sharp drops in May 2026 and peaked at 22 in Jul. 2026



see merged_all_feedback_cleaned.xlsx →Time Series - Negative Ratings by Department (Sheet) for all three Time Series Line Graphs by Department

- The overall trend is very similar to the negative *Ratings* across all *Departments* Times Series (shown above) given that the scale of the Y-axis shows that Services makes up the vast majority of *Ratings* for Kestrel.

Analysis
Overall, negative *Ratings* have been on the gradual rise mostly for the Sales and Services Department for 2026, compared to constant decreases for the Parts Department, possibly indicating improvements in process over time at that Department.

In particular, Services should be focused on for improvement given it that this Department represents the majority of Ratings and that it has the highest proportion of negative Ratings out of all 3 Departments at ~27% and the lowest Proportion of positive Ratings at ~61% (from **Section 2.2** above)

3. Sentiment Analysis on the Comments

- **Purpose:** Reinforce and scrutinise the *Rating/Score* (from 1 to 5) given for a service by Kestrel i.e. Parts, Sales, Service

- Sentiment Analysis on the Comments are to be conducted using Natural Language Processing (NLP) using Python with the help of Claude to classify a Comment of a feedback as either: {Negative, Okay, Positive}

3.1 Sentiment Classification Setup

Task 1: Outline the NLP problem to be solved to Claude with the Comments in the *merge_all_feedback_cleaned.csv* dataset and ask it for help on developing a Sentiment Analysis Python script to classify a Comment as either: {Negative, Okay, Positive}

Prompt 1: *I'm quite rusty with performing Sentiment Analysis with Natural Language Processing using Python. Take this .csv file and help me perform Sentiment Analysis specifically on the 'Comment' column to classify each row as either: Negative, Okay or Positive.*
Note: Each row is a feedback review from a customer.

Here are some suggestions from my experience:

- *Generate a lexicon of words that convey Negative, Okay or Positive sentiment as a reference.*

- *Look into using the Python modules: NLTK (Natural Language Toolkit) and spaCy*

But push back if these suggestions are unnecessary or not required for the problem at hand and try not to rely solely on the 'Ratings/Score' column either.

Then, add the sentiment classification to a new column called 'Sentiment from Comment' and provide an updated .csv file called merged_all_feedback_cleaned_withSentAnaly.csv Do not make any other changes to the .csv file.

Finally, provide the Python code used to perform the operations.

Summary of Response:

- *Agreed on the use of a Lexicon needed to create a custom one because a generic one wouldn't have been able to capture various emotionally-driven human nuances i.e. all-caps, typos, sarcasm*
- *NLTK and spaCy were not required as they mainly perform tokenising and plain Python regex can handle them.*
- *~5% of Comments contradict their respective Rating/Score*
- *Empty Comments were left with an empty 'Sentiment from Comment' entry*
- *Results to the 'Sentiment from Comment' are a first good pass and should not be treated as Ground Truth due to the predictive nature of this problem*

Output: An updated dataset called *merged_all_feedback_withSentAnaly.csv* in the *combined_datasets* folder

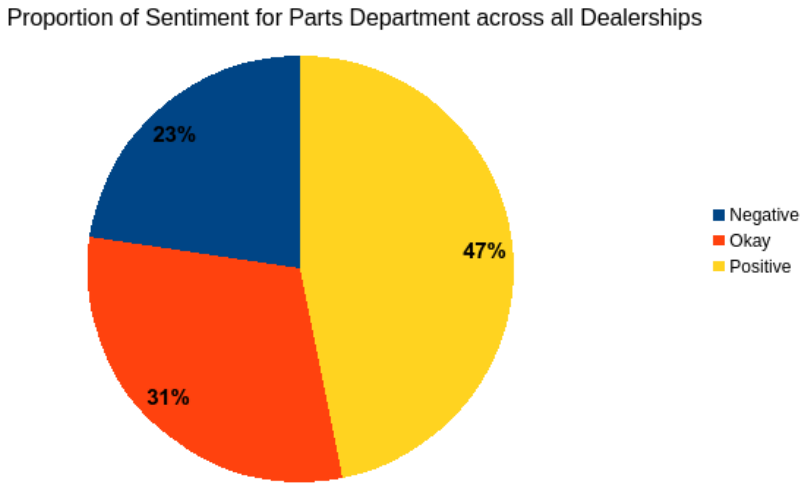
***merged_all_feedback_withSentAnaly.csv* was opened in Excel and a copy was saved as a .xlsx file within the root directory**

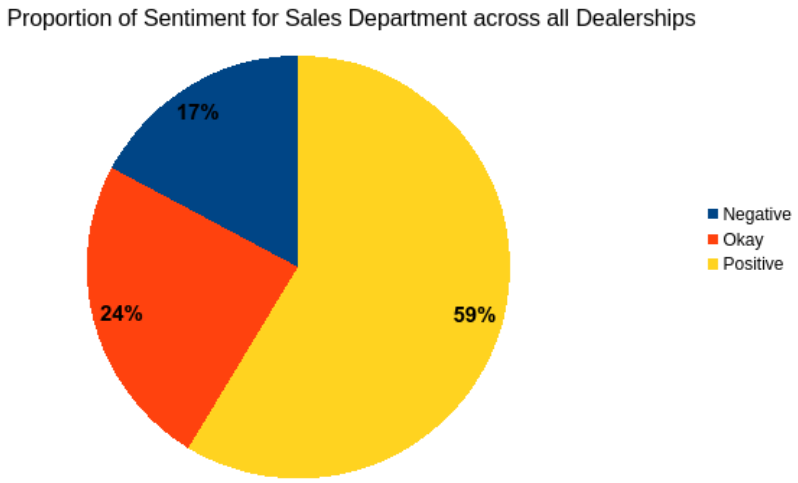
3.2 Pivot Table in Excel to summarise the Sentiment across Dealership Suburbs and Departments

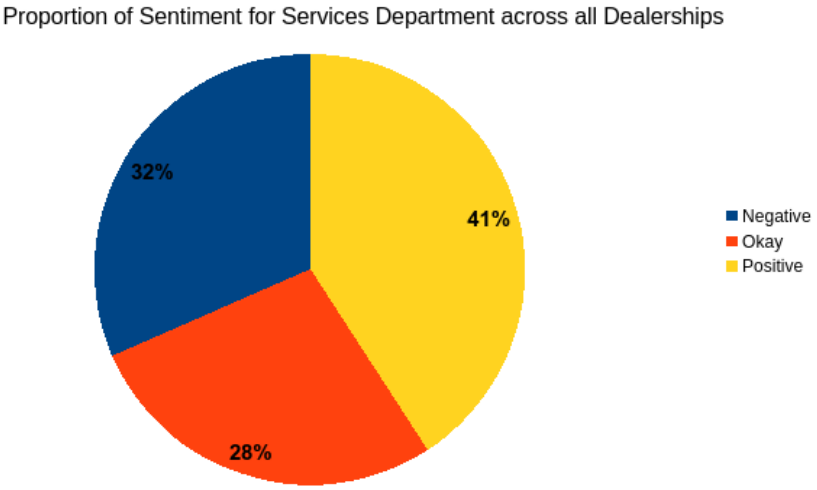
	A	B	C	D	E	F	G	H	I	J	K
1	Count - Sent	Departme	Sentimen	m Comment							
2		Parts			Sales			Service			Total Result
3	Suburb	Negative	Okay	Positive	Negative	Okay	Positive	Negative	Okay	Positive	
4	Ballarat	8	9	10	4	11	26	44	51	64	**227**
5	Bendigo	2	3	6	3	5	11	24	17	32	**103**
6	Box Hill	4	9	12	13	12	42	50	39	61	**242**
7	Chatswood	5	4	17	10	12	26	50	43	54	**221**
8	Chermside	8	6	12	16	24	39	78	77	90	**350**
9	Dandenong	7	8	21	10	21	53	99	70	84	**373**
10	Essendon	6	5	12	11	15	38	67	44	71	**269**
11	Frankston	7	11	5	8	13	29	50	50	60	**233**
12	Geelong	6	7	8	9	9	31	62	42	57	**231**
13	Hunter	7	5	9	6	13	26	46	37	74	**223**
14	Ipswich	3	6	8	8	7	28	58	34	53	**205**
15	Liverpool	3	2	10	19	11	31	35	45	81	**237**
16	Noosa	1		4	1	4	7	1	4	15	**37**
17	North	15	15	26	1	9	3	99	106	145	**419**
18	Parramatta	5	10	15	14	18	52	96	67	85	**362**
19	Penrith	7	16	18	12	15	54	57	58	98	**335**
20	Ringwood	5	10	11	7	19	42	70	69	100	**333**
21	Shepparton		2			3	3	7	4	17	**36**
22	Southport	5	6	5	16	9	26	37	32	55	**191**
23	Springwood	4	6	13	7	17	32	36	41	70	**226**
24	Tamworth		2	1	7	4	10	22	26	30	**102**
25	Toowoomba	4	6	7	1	2	4	54	41	70	**189**
26	Wollongong	3	7	8	6	13	34	40	42	64	**217**
27	**Total Result**	**115**	**155**	**238**	**189**	**266**	**647**	**1182**	**1039**	**1530**	**5361**

see merged_all_feeedback_cleaned_withSentAnaly.xlsx → Pivot Table – Focus on Department (Sheet)

3.3 Pie Charts to convey the proportion of Negative, Okay and Positive Sentiments for difference Departments







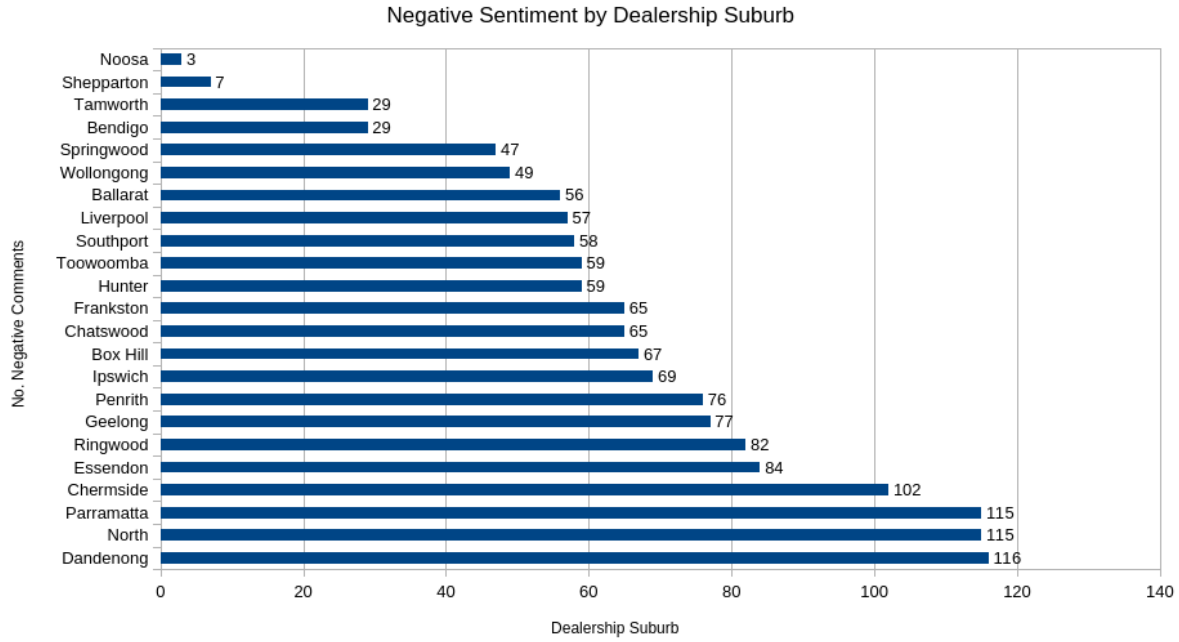
see merged_all_feeedback_cleaned_withSentAnaly.xlsx → Proportion of Negative Sentiment (Sheet) for all three Pie Charts by Department

Comparison of Proportions of Negative and Positive Ratings and Sentiment

Approximate (~) %	**Negative (1 or 2) Ratings**	**Negative Sentiment**	**Difference**	**Positive (4 or 5) Ratings**	**Positive Sentiment**	**Difference**
Parts	16	23	7	72	47	25
Sales	10	17	7	83	59	24
Services	27	32	5	61	41	20

Analysis
Negative *Ratings* and *Sentiment* seem to be much more reliable with a ~5-7% difference. On the other hand, the difference in Positive Ratings and Sentiment is a lot less reliable with a difference ranging from ~20-25%. Nevertheless, it is clear from the table above that Services is in most need of improvement.

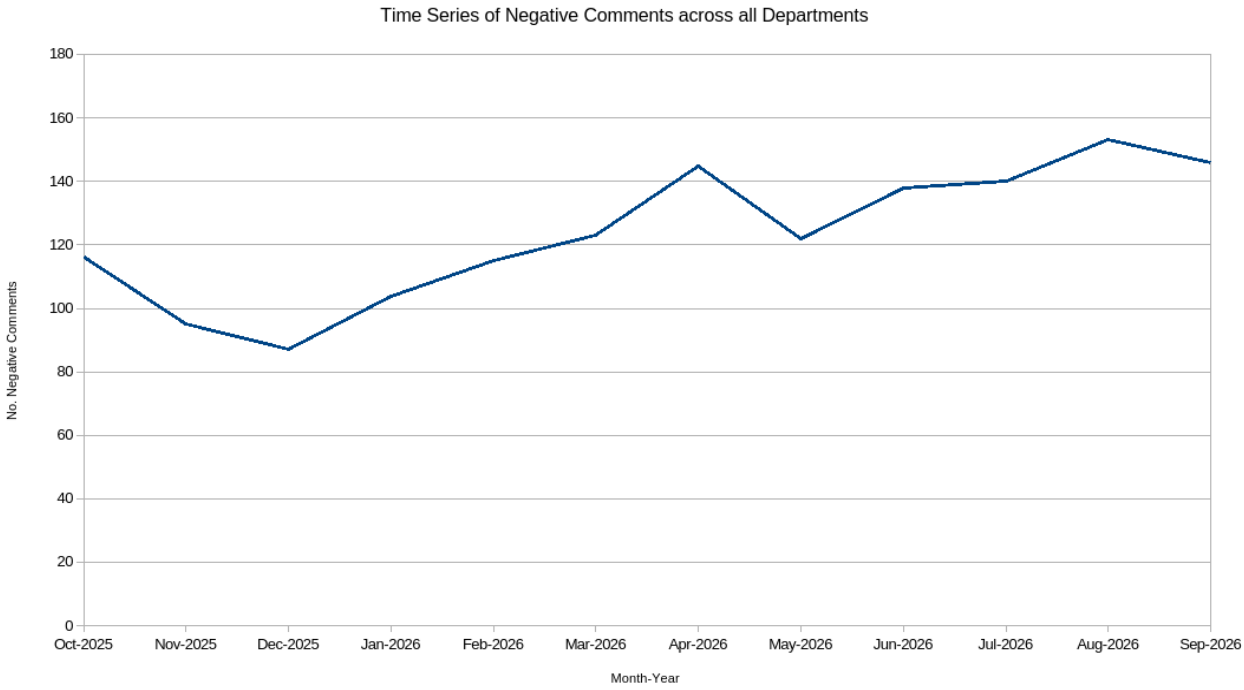
3.4 Bar Chart to show the number of Negative Comments across Dealership Suburbs



see merged_all_feedback_cleaned_withSentAnaly.xlsx → Pivot Table – Focus on Dealership (Sheet)

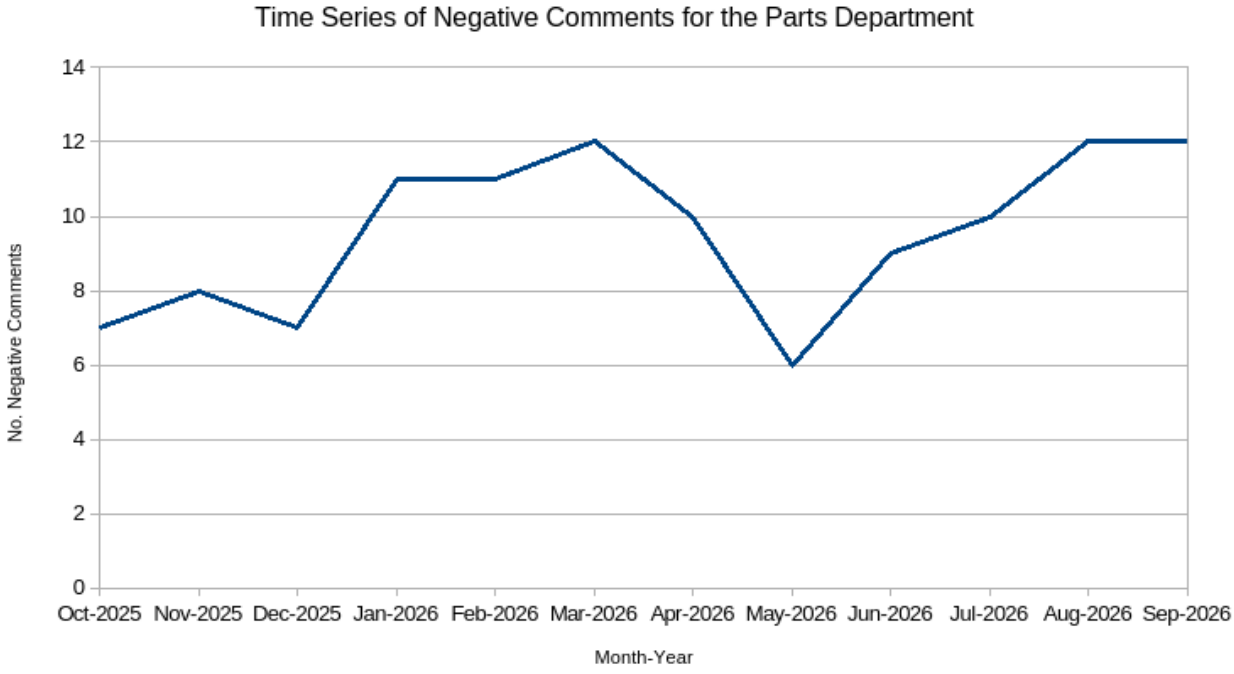
- The Trend and Rankings of Dealership Suburbs from least Negative Comments to most Negative Comments are very similar to the Bar Chart in **Section 2.4** above
    - As such, the insights drawn are basically the same
- Median Negative Comments: **65** compared to Median Negative Ratings: **67** (from **Section 2.4** above) between the 23 Dealership Suburbs is very close.

3.5 Time Series of Negative Comments over time

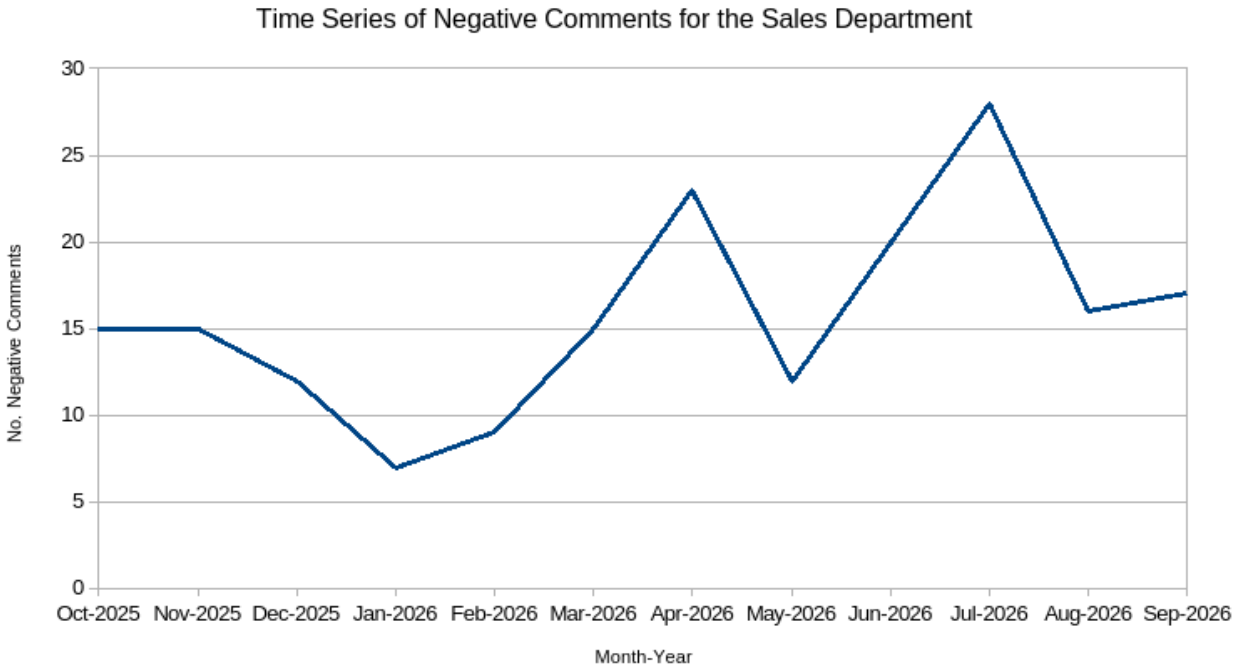


see merged_all_feedback_cleaned_withSentAnaly.xlsx → Time Series – Negative Comments for all Departments

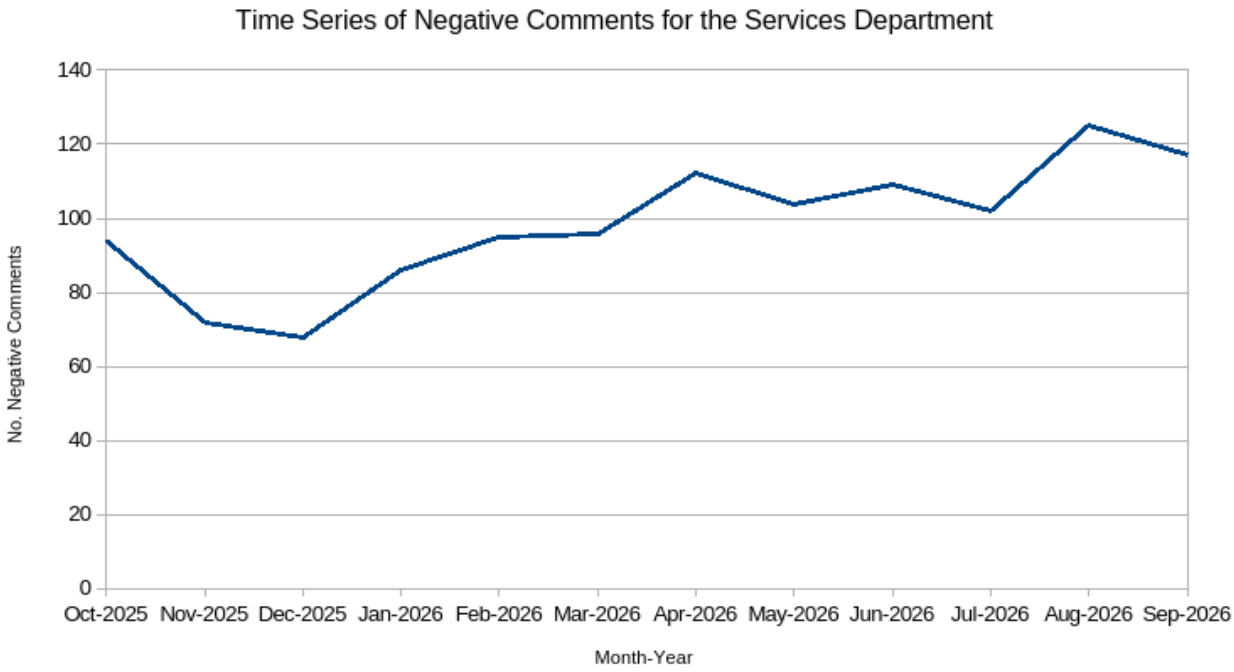
- Negative *Comments* over time follow a near identical trend to negative *Ratings* over time (from **Section 2.5** above).
    - The insights drawn are basically the same, negative Comments are on the rise throughout most of 2026 since its lowest point during Dec. 2025 which then peaked during Aug. 2026



- A cyclical trend is shown, increasing until a peak of 12 during Mar. 2026, a strong decrease then increase again to the same peak of 12 during Aug. and Sep. 2026



- An initial gradual decline of negative *Comments* for the Sales *Department* until Jan. 2026 where negative *Comments* were on the rise throughout 2026



see merged_all_feedback_cleaned_withSentAnaly.xlsx → Time Series – Negative Comments by Department (Sheet) for all three Times Series of Negative Comments by Department

- An initial gradual decline of negative *Comments* for the Services *Department* until Dec. 2025 where negative *Comments* were on the rise throughout 2026

Analysis

- The trend of negative *Comments* for the Parts *Department* does not seem to match the trend taken of negative *Ratings* (from **Section 2.5** above).

    - Perhaps, unhappy customers who bought parts were more likely to leave either a Negative rating or Negative comment but not both.
      Given that *Ratings* are more definitive, the Time Series of Negative Ratings for the Parts Department would be a more accurate reflection of customer sentiment for this Department

    On the other hand, the trends of negative *Comments* and *Ratings* for Sales and Services are very similar.

- The Classification of Sentiment may be yielding some inaccuracies as warned by Claude which may be appearing in the Comments for the Parts Department.

- Just like with negative *Ratings* (in **Section 2.5** above), Sales and Services should be focused on for improvement.

4. Answers for Business Questions

Main Drivers of Negative Feedback:

Poor Services are the main drives of Negative Feedback especially at the Parramatta, Dandenong, Chermside and Penrith Dealerships.

Changes over the Year of 2026:

Negative *Ratings/Scores* and *Comments* of Sales and Services have been gradually increasing throughout 2026. The majority of support should be directed towards Services given that this Department makes up the vast majority of their customer feedback and by extension, most likely Kestrel's revenue stream.

Where should Kestrel's focus first, and roughly what is it worth?

Kestrel's focus should be on enacting improvement processes to their Services Department and providing better support and training to the the Dealership Suburbs in Parramatta, Dandenong, Chermside and Penrith. It is worth a lot to invest in better support and training for Services staff in the long-term so that customers return for future business.

Conclusion

Kestrel's business model seems to be highly reliant on their Services Department from having the vast majority of customer feedback yet has the highest proportion of Negative Feedback and lowest proportion of Positive Feedback compared to the other two Departments. Direct support for improvement for Dealerships should be directed to Parramatta, Dandenong and Chermside which each have almost double the Median number of Negative feedback. Lastly, further investigation should be conducted into the Noosa Dealership Suburb regarding why *Ratings* and *Comments* are missing.